

Assignment:-5

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Subject: Operating system

① What are the three main techniques for performing I/O? Briefly explain each.

- • Programmed I/O: CPU continuously checks device
- Interrupt-driven I/O: CPU works, interrupted when I/O completes
- DMA: Direct transfer between memory & device (no CPU).

② Explain the role of Interrupts in modern operating systems. Draw the interrupt-driven I/O cycle.

→ Interrupts: Inform CPU when I/O finishes → avoids waiting

cycle:

Request → CPU continues → Interrupt → ISR executes → resume

③ What is DMA? Describe the steps involved in a DMA transfer. How does it free up the CPU?

→ DMA: Direct memory Access (no CPU during transfer)

Steps:

- 1) CPU sets DMA
- 2) DMA transfer data
- 3) Interrupt on completion

Benefit: CPU free for other work.

④ Define the following terms related to disk structure: tracks, sector, cylinders, platters.

- Tracks: circular path
- sector: smallest unit
- Cylinders: same track across platters
- Platters: disk surface

⑤ Explain why disk scheduling is important.
List any three disk scheduling algorithms.

→ Importance: Reduces seek time, improve efficiency.

Algorithms:

- FCFS
- SSTF
- SCAN

⑥ What is the SSTF disk scheduling algorithm?
What is its main disadvantage?

→ SSTF: closest request served first

Disadvantages: Starvation.

UG
Net
⑦ A disk queue has request for cylinders:
98, 183, 37, 122, 14, 124, 05, 67. The
head is at 53. Calculate the total head
movements for FCFS.

→ Movement =

53 → 98 (45)

98 → 183 (85)

183 → 37 (146)

34 $\rightarrow$ 122 (85)

122 $\rightarrow$ 14 (208)

14 $\rightarrow$ 122 (110)

124 $\rightarrow$ 65 (59)

65 $\rightarrow$ 67 (2)

$\therefore$ Total = 640 cylinders

⑧ For the same disk queue and head at 53 moving towards larger numbers, calculate total head movement for SCAN.

$\rightarrow$ Order:

53 $\rightarrow$ 65 $\rightarrow$ 67 $\rightarrow$ 98 $\rightarrow$ 122 $\rightarrow$ 124 $\rightarrow$ 183 $\rightarrow$ end $\rightarrow$ 34 $\rightarrow$ 14

$\therefore$ Total $\approx$ 337 cylinders.

⑨ Differentiate between SCAN and C-SCAN disk scheduling algorithms. which one provides more uniform wait time and why?

$\rightarrow$ • SCAN: moves back & forth

• C-SCAN: one direction only

* C-SCAN: better (uniform wait time)

⑩ Explain the concept of Disk Formatting.
What is the difference between low-level and high-level formatting?

→ Disk Formatting: Prepares disk for use

- Low-level : creates sectors / tracks
- High-level : creates file system.

⑪ What is Swap space? where is it typically located, and what is its purpose?

→ Swap Space: Disk space used as extra memory

Location: Hard disk / SSD

Purpose: Supports virtual memory.

⑫ What is a RAID? List any three goals of RAID.

→ RAID : combines multiple disks.

Goals:

- Reliability
- Performance
- Fault tolerance

⑬ Explain RAID level 0 and RAID level 1. Compare them in terms of performance and fault tolerance.

- RAID 0: striping → fast, no protection
- RAID 1: mirroring → safe, slower

⑭ Explain RAID level 5. How does it differ from RAID level 4?

→ RAID 5: Distributed parity

Difference:

- RAID 4 → single parity disk
- RAID 5 → parity spread across disks

⑮ What is the difference between a boot block and a bad block? How does the OS handle a bad block?

- Boot Block: contains boot code
- Bad Block: Damaged sector

Handling:

- Mark unstable
- Remap to spare sector

16.a) Explain in detail the various layers of the I/O software. What is the purpose of each layer?

- User-level → interface
 - Device-independent → common services
 - Drivers → device control
 - Hardware → actual I/O

b) Compare programmed I/O, interrupt-driven I/O, and DMA. When would you use each?

- Programmed → simple, small tasks
- Interrupt → moderate I/O
- DMA → large data transfers

17.a) Calculate total seek distance (SSTF, SCAN, C-SCAN, LOOK).

→ SSTF ≈ 236

SCAN ≈ 331

C-SCAN ≈ 382

LOOK ≈ 299

b) Which algorithm performed best?

→ SSTF

18 a) Describe structure of modern hard disk & timings.

→ Structure: platters, heads, arm, spindle

Timings:

- Seek time → head movement
- Rotational latency → wait for sector
- Transfer time → data read

b) Calculate average time to read 512-byte sector.

→ 7200 RPM - 8.33 ms/rotation

Avg latency ≈ 4.16 ms

seek ≈ 8 ms

Transfer negligible

$\therefore$ Total ≈ 12.9 ms

19. a) Explain RAID level (0, 1, 4, 5).

→ RAID 0 → striping, no safety

• RAID 1 → mirror, safe

• RAID 4 → single parity

• RAID 5 → distributed parity

b) RAID 5 failure recovery?

- Use Parity + remaining disks
- Reconstruct lost data.

20. a) Explain booting process (MBR, boot loader, bootstrap).

→ MBR: First sector, starts boot

Bootloader: Loads OS

Bootstrap: Initializes system.

b) Swap space management techniques & trade-offs.

→ Separate Partition → fast

- File → flexible

Trade off: speed vs flexibility

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